

7.4 Trigonometric Ratios on the Coordinate Plane

Lesson Introduction

 Benchmarks	MA.912.T.1.1 Define trigonometric ratios for acute angles in right triangles.		MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.		 Learning Target	I can determine trigonometric ratios using the coordinate plane.
 Benchmark Coherence	Building On			Working Toward		
	MA.8.GR.1.1 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.			N/A		
 Essential Questions	- How are trigonometric ratios related in similar triangles? - How do I find trigonometric ratios using a coordinate plane?					
 Lesson Narrative	Through work on the coordinate plane, students discover the relationship between trigonometric ratios in similar triangles. Using the knowledge that trigonometric ratios are equivalent in similar triangles, students will find trigonometric ratios of given reference angles.					
 Component Summary	7.4.1 Warm-Up (~5 min.)	Describe a pattern (<i>SE p. 388</i>)	7.4.3 Guided Instruction (10-15 min.)	Approximate trigonometric ratios (<i>SE p. 390</i>)		
	7.4.2 Collaboration (10-15 min.)	Define trigonometric ratios on the coordinate plane (<i>SE p. 388</i>)	7.4.4 Guided Practice (5 min.)	Find trigonometric ratios (<i>SE p. 392</i>)		
	7.4.5 Wrap-Up (~5 min.)	Explain the relationship between trigonometric ratios and similar triangles (<i>SE p. 393</i>)				
 Resources	Glossary		Materials		Additional Preparation	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Sine - Cosine - Tangent		- Scientific calculator (optional) Colored Pencils (optional) Highlighters		N/A	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may flip the answers for sine and cosine if they have mislabeled the sides of a right triangle. - Students may forget to take the square root when using the Pythagorean Theorem to solve. - Students may divide by 2 when using the Pythagorean Theorem to solve, instead of taking the square root.	Consider allowing students to trace triangles on patty paper if they are struggling with solving triangles that are overlapping.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share After completing the chart with the trigonometric ratios in 7.4.2 Collaboration, have the students pause to allow them to ponder the question, “What similarities do they notice?” Students may then work with a partner to answer the question and share with the class or surrounding groups.	MA.K12.MTR.2.1: Multiple Representations Throughout the lesson, spotlight the different ways trigonometric ratios can be represented through similar triangles. Consider using the language embedded in the benchmark clarifications for MA.K12.MTR.2.1 when facilitating discussion.
 Differentiate Instruction	Using color coding methods when working with similar triangles will help visual students keep their work organized and draw conclusions from their findings in the collaboration component of this lesson. The 7.4.2 Collaboration allows entry points for auditory learners to discuss and share their findings with a partner, small group, or the class as a whole. Additional differentiation opportunities are denoted throughout the lesson guide.	
Lesson Notes:		

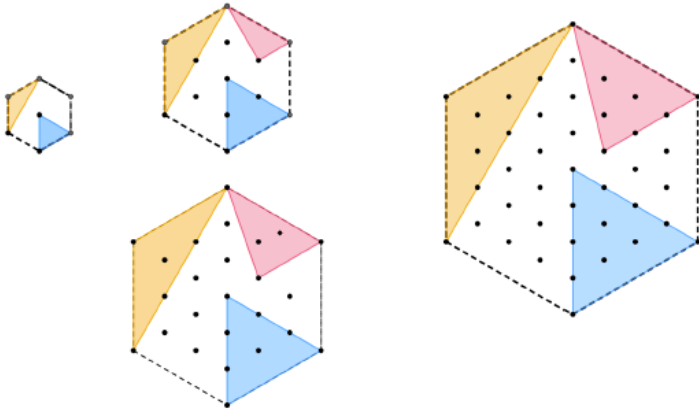
7.4.1 Warm-Up: Patterns

Component Overview: Students are observing and stating the patterns seen in the images given.



7.4.1 Warm-Up: Patterns

Four hexagons are shown.



For a different entry point for students who struggle writing, provide the next two blank hexagons with the dots for students to draw the 5th and 6th patterns. Then have students verbally state how they knew to draw the patterns.

1. Describe the pattern of each triangle.

Triangle	Pattern Description
Yellow	<i>The triangle appears to be isosceles. In each shape in the sequence, the sides seem to be increasing by 1 unit dot each time.</i>
Red	<i>The triangle appears to be scalene. In each shape in the sequence, the sides seem to be increasing by 1 unit dot each time.</i>
Blue	<i>The triangle appears to be equilateral. In each shape in the sequence, the sides seem to be increasing by 1 unit dot each time.</i>

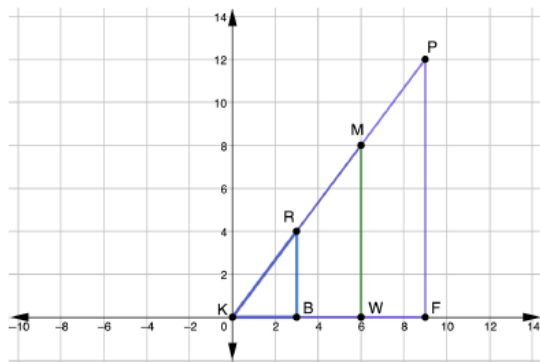
7.4.2 Collaboration: Defining Trigonometric Ratios

Component Overview: Through collaboration with peers, students are working together to discover the relationship between trigonometric ratios and similar triangles on the coordinate plane.



7.4.2 Collaboration: Defining Trigonometric Ratios

- Three right triangles are shown on the coordinate grid.



- Discuss with your partner how to justify that ΔKBR , ΔKWM , and ΔKFP are similar. Summarize your discussion.

Answers will vary. The three triangles are similar by AA similarity because they share $\angle K$ and they all have a right angle. Also, the sides are proportional.

- Complete the table.

Segment	Length	Segment	Length
$\overline{KB}$	$KB = 3$	$\overline{WM}$	$WM = 8$
$\overline{BR}$	$BR = 4$	$\overline{KF}$	$KF = 9$
$\overline{KW}$	$KW = 6$	$\overline{FP}$	$FP = 12$



Consider selecting students who used different methods and proved similarity correctly to discuss why each method is correct. If all students used the same method, consider asking students to prove similarity using a different method.

7.4.2 Collaboration: Defining Trigonometric Ratios (continued)

- c. Use the lengths from the table to determine each. Show your work.

Segment	Length
$\overline{KR}$	$3^2 + 4^2 = (KR)^2$ $9 + 16 = (KR)^2$ $25 = (KR)^2$ $\sqrt{25} = KR$ $5 = KR$
$\overline{KM}$	$6^2 + 8^2 = (KM)^2$ $36 + 64 = (KM)^2$ $100 = (KM)^2$ $\sqrt{100} = \sqrt{(KM)^2}$ $10 = KM$
$\overline{KP}$	$9^2 + 12^2 = (KP)^2$ $81 + 144 = (KP)^2$ $225 = c^2$ $\sqrt{225} = \sqrt{KP^2}$ $\sqrt{225} = KP$ $15 = KP$

- d. Using $\angle K$ as the reference angle, find the three ratios for each triangle.

Triangle	Cosine	Sine	Tangent
$\triangle RKB$	$\frac{3}{5}$	$\frac{4}{5}$	$\frac{4}{3}$
$\triangle MKW$	$\frac{6}{10}$	$\frac{8}{10}$	$\frac{8}{6}$
$\triangle PKF$	$\frac{9}{15}$	$\frac{12}{15}$	$\frac{12}{9}$



For a visual approach, have students highlight the triangle segments in different colors.



Common Misconception
Students may incorrectly solve using the Pythagorean Theorem by not taking the square root to find the hypotenuse.



Think-Pair-Share
After completing the chart with the trigonometric ratios, allow students to ponder the question, "What similarities do you notice with the ratios?" Students may then work with a partner to answer the question and share with the class or surrounding groups.

7.4.2 Collaboration: Defining Trigonometric Ratios (continued)

2. $\triangle HWB$ is graphed on the coordinate grid, where $m\angle WBH = 90^\circ$. The coordinates of the vertices are $H(-4, 5)$, $W(5, -3)$, and $B(-4, -3)$.
- a. Discuss with your partner how to use the coordinates to determine the trigonometric ratios. Summarize your discussion.
Answers will vary.
I can use the distance between the points to determine the length of the sides. Then, using these lengths, I can determine the trigonometric ratios.
- b. Complete the table. Show your work.

Segment	Length
$\overline{WB}$	$WB = 9$
$\overline{BH}$	$BH = 8$
$\overline{HW}$	$HW = \sqrt{145}$

- c. Using $\angle WHB$ as the reference angle, find the three ratios.

Cosine	Sine	Tangent
$\frac{8}{\sqrt{145}}$	$\frac{9}{\sqrt{145}}$	$\frac{9}{8}$



Consider providing struggling students graph paper to use as a visual to help them conceptualize using coordinates to determine the trig ratios.



Consider determining how you would like students to answer the distance of HW .



Common Misconception

If students find that $HW = 72.5$, then students divided by 2 instead of taking the square root. Consider facilitating a discussion with the class or a small group of students about how powers and radicals are inverse operations.

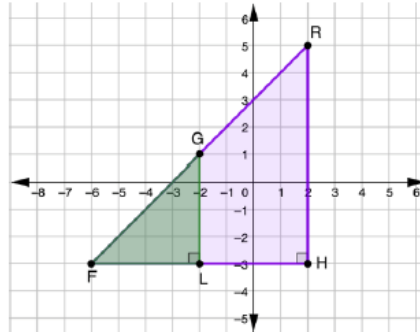
7.4.3 Guided Instruction: Using Trigonometric Ratios

Component Overview: The guided instruction works through how to find trigonometric ratios and how to approximate the value of the ratio.



7.4.3 Guided Instruction: Using Trigonometric Ratios

1. $\triangle FLG$ and $\triangle FHR$ shown on the coordinate grid are isosceles right triangles.



- a. Determine the measure of each angle.

Angle	Measure
$\angle LFG$	45°
$\angle FGL$	45°
$\angle FRH$	45°

- b. Using $\angle F$ as the reference angle, find the three ratios for each triangle.

Triangle	Cosine	Sine	Tangent
$\triangle FLG$	$\frac{4}{\sqrt{32}}$	$\frac{4}{\sqrt{32}}$	$\frac{4}{4} = 1$
$\triangle FHR$	$\frac{8}{\sqrt{128}}$	$\frac{8}{\sqrt{128}}$	$\frac{8}{8} = 1$

- c. Use your calculator to find each. Round to the nearest thousandth, if necessary.

Angle	Cosine	Sine	Tangent
45°	0.707	0.707	1

- d. How are the ratios you found using $\angle F$ as the reference angle related to the values you found for the ratios of a 45° angle?

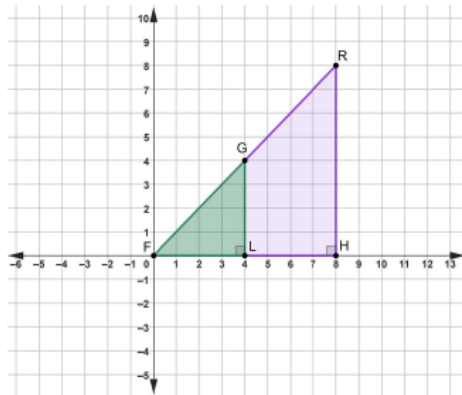
The values found using the calculator for a 45° angle are equivalent to the decimal approximations of the ratios found using $\angle F$ as a reference angle.



- What does it mean for a triangle to be an isosceles triangle?
- How can you find the base angles in a right isosceles triangle?

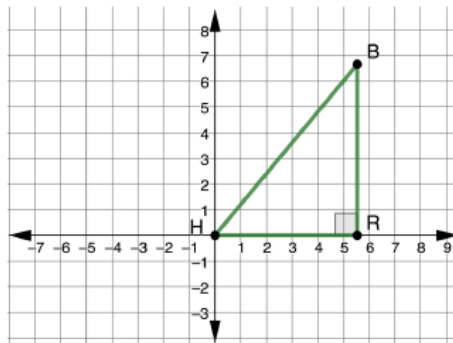
7.4.3 Guided Instruction: Using Trigonometric Ratios (continued)

- e. $\triangle FLG$ and $\triangle FHR$ were moved on the coordinate grid so that F is at the origin.



Work with your partner to determine whether or not the values of the trigonometric ratios will change because the triangle moved. Summarize your findings. *The values will not change, for the figure was only repositioned by a rigid transformation and not resized.*

2. $\triangle HBR$ is shown on the coordinate grid, where
 $m\angle HRB = 90^\circ$,
 $m\angle RHB = 50^\circ$,
 $HR = 5.567$, and
 $HB = 8.66$.



- a. Explain why the ratio $\cos(50^\circ) \approx \frac{5.567}{8.66}$ is true.

Answers may vary.

In a right triangle, the cosine of an angle is the ratio of $\frac{\text{adjacent}}{\text{hypotenuse}}$. In $\triangle HBR$, the length of the side adjacent to $\angle BHR$, which has the measure of 50° , is 5.567 and the hypotenuse of the triangle is $\overline{HB}$, which has a length of 8.66.



Common Misconception

If students find cosine and sine are 1.41, then students used the reciprocal of the trigonometric ratios. This misconception can be addressed by discussing the ratios for each trigonometric function, as well as how to correctly label a right triangle based on a reference angle.



When using a calculator, consider reminding students to make sure their calculator is in degrees, not radians.

7.4.3 Guided Instruction: Using Trigonometric Ratios (continued)

- b. Find each value using a calculator. Round to the nearest thousandth.

$$\cos(50^\circ) = 0.643 \qquad \frac{5.567}{8.66} = 0.643$$

- c. Discuss with your partner why the approximation symbol was used in $\cos(50^\circ) \approx \frac{5.567}{8.66}$. Summarize your thoughts below.

The value of both $\cos 50^\circ$ and $\frac{5.567}{8.66}$ are approximately the same. Using a calculator to approximate each shows that $\cos 50^\circ$ is 0.6427876097 and $\frac{5.567}{8.66}$ is 0.6428406467. These two values are approximately the same and both have the same value if rounded to the thousandths.

- d. Work with your partner to find $\sin(H)$. Round to the nearest thousandth.

$$5.567^2 + (RB)^2 = 8.66^2$$

$$RB \approx 6.634$$

$$\sin(H^\circ) = \frac{6.634}{8.66}$$

$$\sin(H^\circ) \approx 0.766$$

- e. Use your calculator to find $\sin(50^\circ)$.

$$\sin(50^\circ) \approx 0.766$$

- f. Explain why $\sin(50^\circ) \approx \sin(H)$.

Both of these values when rounded equal approximately 0.766 so they are approximately equal to each other.

- g. Work with your partner to write the coordinates of each vertex.

Vertex	Coordinates
R	(5.567, 0)
B	(5.567, 6.634)



Monitor students as they discuss with their partner, and identify pairs of partners whom you wish to share with the class.

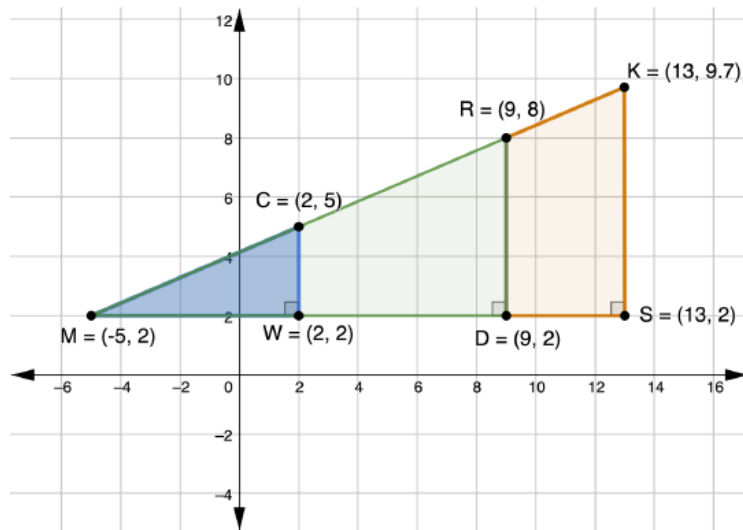
7.4.4 Guided Practice: Trigonometric Ratios and Similar Triangles

Component Overview: Through this teacher-led practice, students will practice approximating trigonometric ratios.



7.4.4 Guided Practice: Trigonometric Ratios and Similar Triangles

1. $\triangle CMW$, $\triangle RMD$, and $\triangle KMS$ are shown on the coordinate grid.



Using $\angle M$ as the reference angle, find the three ratios for each triangle. Write the values as a decimal rounded to the nearest thousandth.

Triangle	Cosine	Sine	Tangent
$\triangle CMW$	0.919	0.394	0.429
$\triangle RMD$	0.919	0.394	0.429
$\triangle KMS$	0.919	0.394	0.429

2. Yasmeen said she only found the ratios for $\triangle CMW$ then applied them to the other triangles. Explain using mathematics why she can do that.

All three triangles are similar by AA similarity therefore all of their sides are proportional.



Allow informal observation throughout the first two activities to dictate how “guided” this guided practice should be for you and your classroom. This could mean small group instruction for some while others attempt independently, or it could mean teacher facilitation of practice with students actively working as a whole group.



For a kinesthetic approach, consider allowing students to copy the three triangles on to separate graphs. Students can label the different side lengths accordingly.



Common Misconception

Monitor student answers to check that sine and cosine are not being switched. If they are switched, it may indicate students have mislabeled the sides. Consider facilitating a discussion about how to label the sides of the triangle as opposite, adjacent, and hypotenuse.

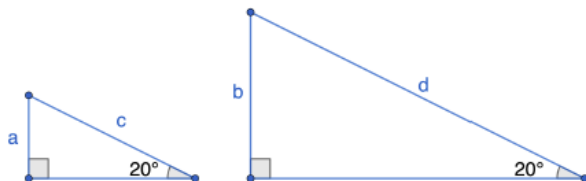
7.4.5 Wrap-Up: Ticket Out the Door

Component Overview: Students will complete a Ticket Out the Door that demonstrates their understanding of the relationship between trigonometric ratios and similar triangles.



7.4.5 Wrap-Up: Ticket Out the Door

1. Two triangles are shown.



Use student responses along with teacher observations to plan for remediation and future lessons.

- a. Describe how the ratios $\frac{a}{c}$ and $\frac{b}{d}$ are related.

$\frac{a}{c} = \frac{b}{d}$ because the triangles are similar and therefore their sides are proportional.

- b. What mathematical relationship(s) did you use to complete your description?

Answers may vary. Since the triangles are similar by the AA theorem, the ratios will simplify to equivalent values.